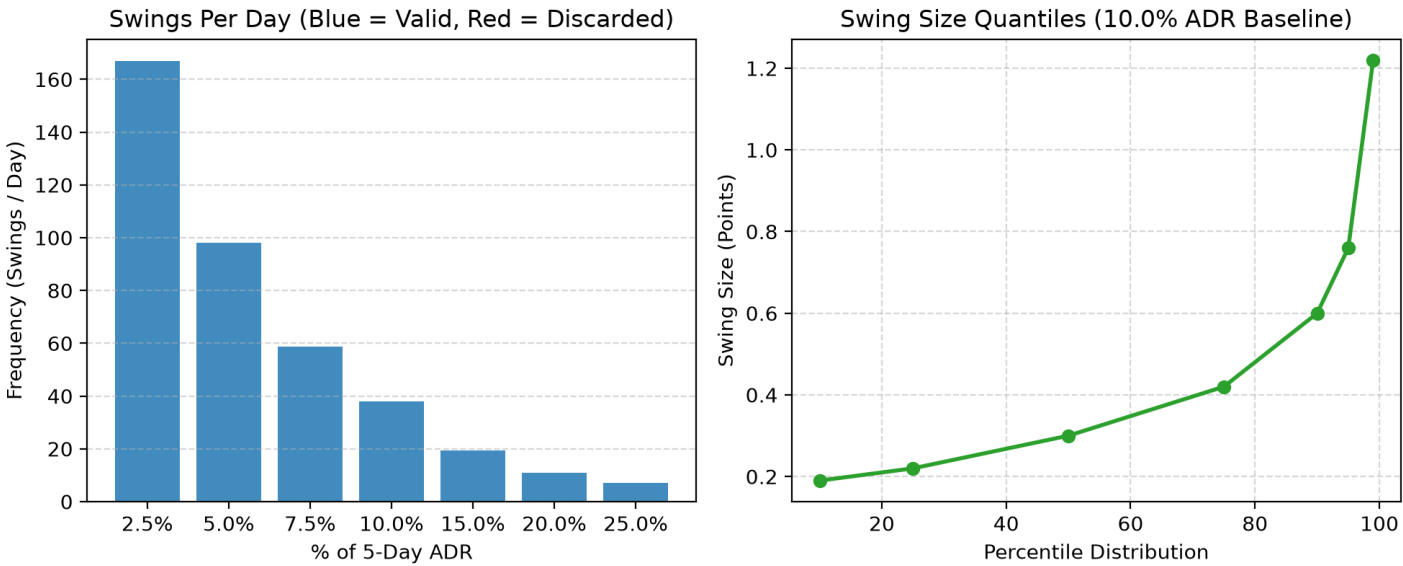


Statistical Swing Analysis Report: VALE3 (1min)

5-Day Average Daily Range (ADR_5d): 1.66 pts

VALE3 (1min) - Swing Distribution & Sensitivity Analysis



Parameter Sweep & Noise Filtering Matrix

Pct ADR	Threshold Pts	Total Swings	Swings Per Day	Avg Bars Per Swing	Mean Size	Median Size	P75 Size	P90 Size	P95 Size	Status
2.5%	0.04	19532	166.9	3.5	0.15	0.12	0.18	0.27	0.36	VALID
5.0%	0.08	11466	98.0	5.2	0.21	0.17	0.25	0.37	0.45	VALID
7.5%	0.12	6863	58.7	8.0	0.29	0.23	0.34	0.47	0.6	VALID
10.0%	0.17	4432	37.9	11.8	0.36	0.3	0.42	0.6	0.76	VALID
15.0%	0.25	2285	19.5	22.0	0.52	0.42	0.61	0.88	1.07	VALID
20.0%	0.33	1288	11.0	38.1	0.7	0.58	0.85	1.17	1.48	VALID
25.0%	0.41	814	7.0	59.7	0.9	0.73	1.07	1.52	1.83	VALID